



Patient Zero.

Reflections on diffusion cascade source detection in undirected networks.

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To what extent can the source of a diffusion process be inferred from the **undirected** cascade structure, and how does prediction accuracy depend on **R_0** ?

STEP 1 · HIDDEN

Contact network & transmission tree



STEP 2 · STRIP DIRECTIONS

Observed undirected subgraph



STEP 3 · RECOVER

Rank every node by $P(\text{source})$



The model only ever sees the right-hand box; the transmission tree is hidden. In a misinformation setting the "source" is a real person, so a wrong answer carries real stakes.

What we built, in one row.

01 · NETWORKS

Two topologies, matched $\langle k \rangle$

Erdős–Rényi and Barabási–Albert, $N = 200$, average degree ≈ 6 .

02 · DIFFUSION

IC across five R_0

$R_0 \in \{0.5, 1, 2, 3, 5\}$.
1,000 cascades per R_0 , all of size 25.

03 · FEATURES

Seven structural signals

Geometric, neighbourhood, and component features on the undirected subgraph.

04 · LEARN

Random Forest, 5 seeds

StratifiedGroupKFold so cascades never leak across folds. Class-balanced.

05 · COMPARE

Four centrality baselines

Jordan centre, closeness, betweenness, degree, plus random as a null-model floor.

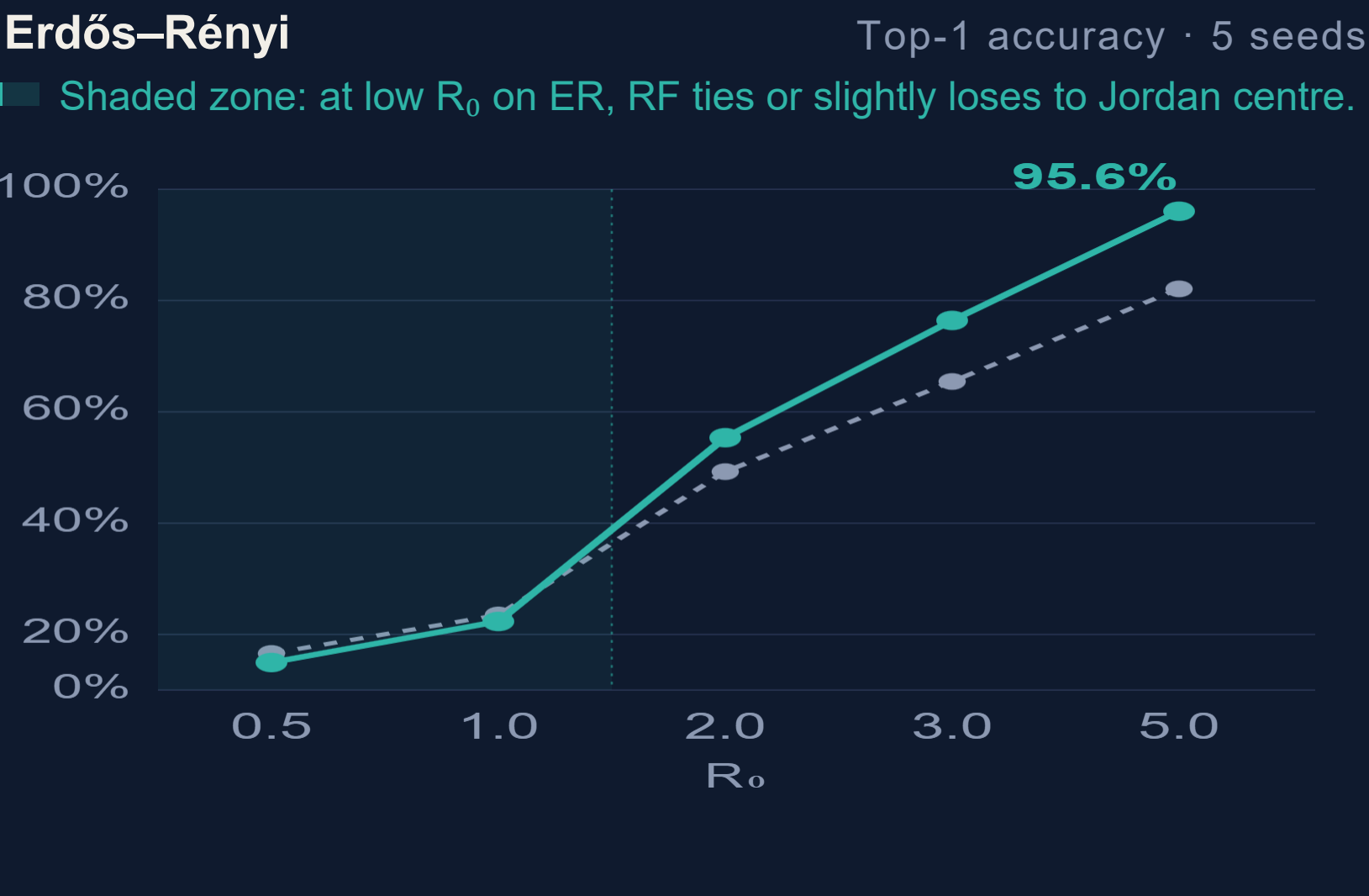
THEN: TRANSFER TEST

Weibo reposts · OutbreakTrees · FalseNews. Three real datasets. Controlled training, then a real-world test for distribution shift.

RF beats the centrality baselines, but the gap depends on topology and R_0 .



BA. RF wins at every R_0 ; 8.5% climbing to 93.5% Top-1

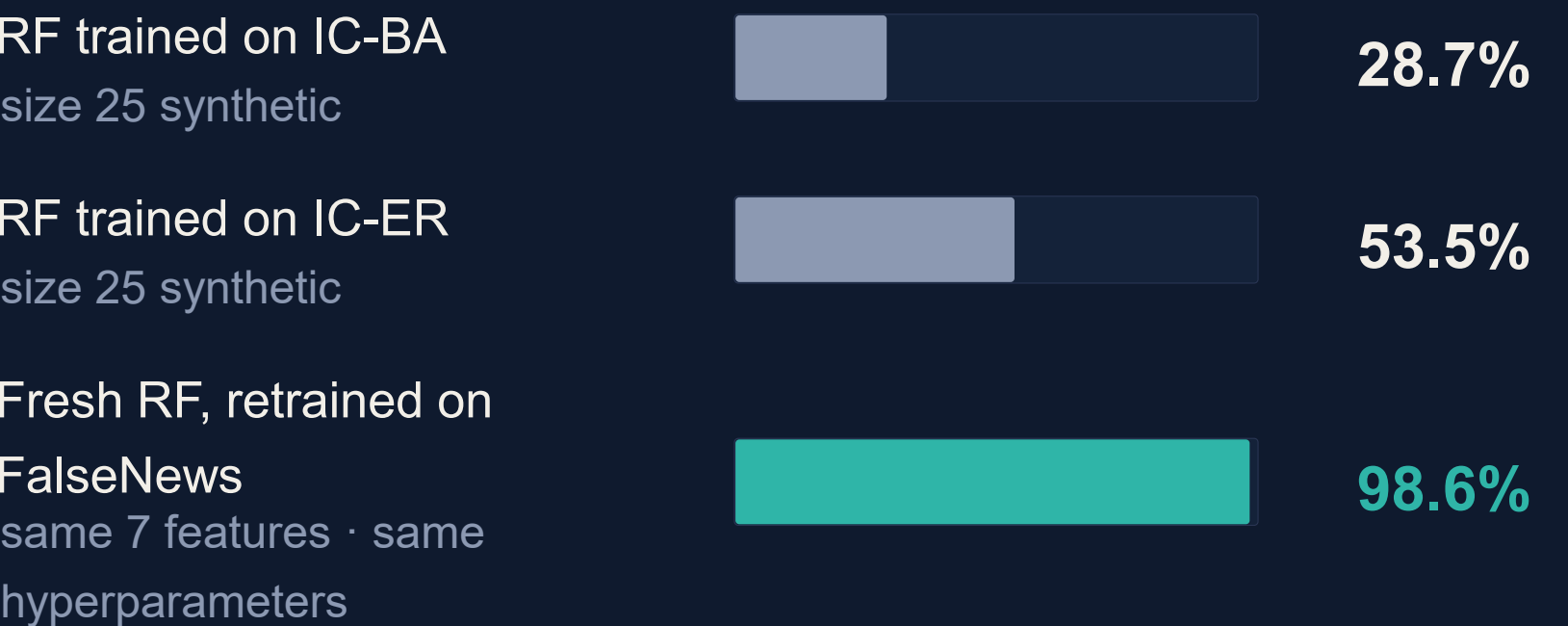


ER. RF wins from $R_0 \geq 2.0$; below that, survivor cascades carry too little signal

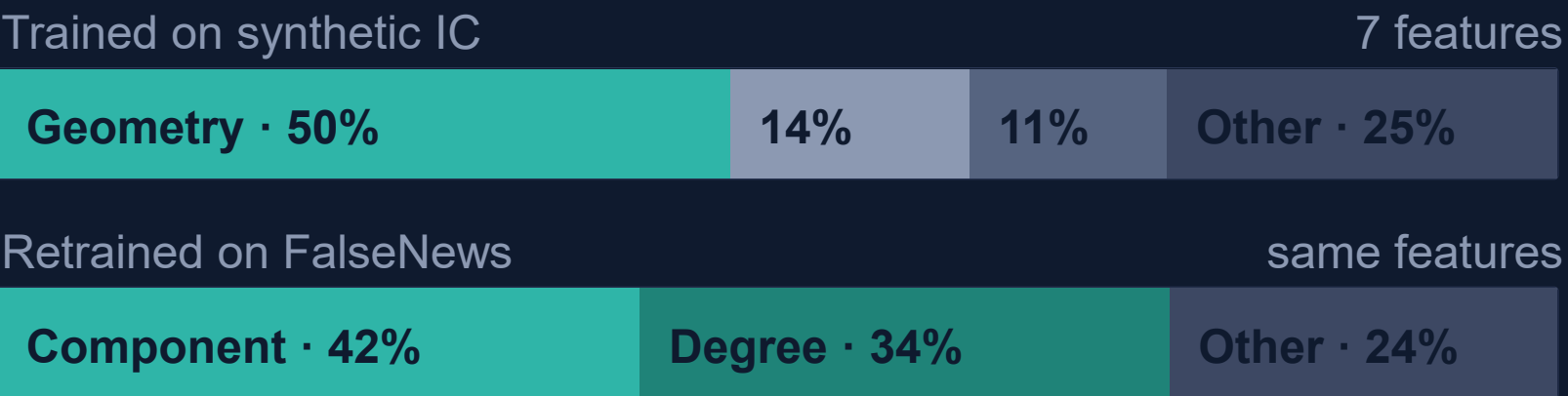
Drivers. Jordan-distance + endpoint-balance $\approx$ 50% of feature importance

Transfer is conditional. The gap isn't the model.

TOP-1 ACCURACY · FALSENEWS (MEDIAN-101 SLICE)



WHERE THE SIGNAL LIVES · FEATURE IMPORTANCE



Same architecture. Different distribution moves the signal from geometry to degree. This is a social-data-bias problem, not a model issue.

Same architecture. Different data. **Different model.** The gap is distribution shift, not model design.

Three course concepts, turned on my own work.

SOUND SCIENCE · BASELINES

Was it a fair comparison?

My RF was tuned over 5 seeds with group-aware CV ; the centrality baselines ran at their defaults. Unequal effort on each side can be considered a victim to the **unfair-comparison** trap. The random floor is my only honest null model, but may result in inflated values.

Bergstrom & West, ch. 10 · "Sound science"

SOCIAL DATA BIAS

The shift is bias, not noise.

The IC→FalseNews drop is **population bias** (synthetic IC assumes uniform, memoryless spreaders) plus **behavioural bias** (real misinformation accounts spread, and are sampled, differently from my simulation).

Olteanu et al. (2019)

CALLING BS ON MYSELF

Is the result actually real?

The geometry→degree flip rests on a 101-cascade slice. Before I believe my own plot, I have to rule out the **clustering illusion**, a pattern in a small sample, with proper significance testing.

Lecture 6

Four limitations, read through the course lens.

CONSTRUCT VALIDITY



Cascade size fixed at 25

Removes size as a confound, but [Juul & Ugander \(2021\)](#) shows size itself shapes structure. We may have over-controlled.

MODELLING ASSUMPTION



Independent Cascade is a strong assumption

Memoryless, one-shot. [Lehmann & Juul \(2022\)](#) argues social-media diffusion is complex contagion, which IC can't express.

SOUND SCIENCE



Only one architecture tested

RF was chosen for interpretability. No GNN baseline means we can't bound the value of representation learning vs hand-engineered features.

REPRODUCIBILITY



Simulator processes neighbours in fixed order

At the size-25 cutoff, whichever spreader iterates first wins. Effect likely small but acknowledged. Needs a randomised-order ablation.

Three directions I'd pursue.

01

Cross-process training

Train on a mixture of IC, SIR, and linear-threshold cascades. Test whether the feature set survives a spreading-process shift, not just an R_0 shift.

02

Size-cutoff ablation

Repeat the R_0 sweep at multiple size cutoffs (15, 25, 50, 100) to disentangle the contribution of cascade dynamics from cascade geometry.

03

GNN baseline

Add a graph neural network to bound the upper limit of representation learning, while weighing its higher energy cost against the accuracy gain (sustainability).

Undirected source detection is possible, but **conditional**. On enough branching, on useful topology, and on a training distribution that matches deployment.

SHOULD IT BE DEPLOYED?

Not as a public oracle. At $R_0 \leq 1$ it barely beats chance, and treating node degree as a proxy for the source would wrongly flag just based off being a popular reposter. That is the platform bias (lecture 7) I would be cautious of before deployment. Although this is a real relationship the model should be picking up, it should do so carefully.

AT WHAT COST?

The GNN that might close the gap also raises test-time compute. Accuracy that doesn't justify its energy isn't automatically the sustainable choice.

Thesis conclusion, reflected through Reflections on Data Science